Flexible $p(\chi_{\rm eff}, \chi_p \mid m_1, q)$

all q | $q \in [0.8, 1.0]$ | $q \in [0.3, 0.65]$

all m_1 | $m_1 \in [8, 12]\,M_\odot$ | $m_1 \in [16, 22]\,M_\odot$ | $m_1 \in [30, 40]\,M_\odot$ | $m_1 \in [55, 65]\,M_\odot$

$P(q \in Q)$ $0.6^{+0.19}_{-0.42}$ | $P(q \in Q)$ $0.16^{+0.16}_{-0.077}$

$P(m_1 \in M)$ $0.66^{+0.093}_{-0.13}$ | $P(q \in Q \mid m_1 \in M)$ $0.72^{+0.21}_{-0.6}$ | $P(q \in Q \mid m_1 \in M)$ $0.055^{+0.21}_{-0.05}$

$P(m_1 \in M)$ $0.069^{+0.05}_{-0.032}$ | $P(q \in Q \mid m_1 \in M)$ $0.076^{+0.21}_{-0.067}$ | $P(q \in Q \mid m_1 \in M)$ $0.65^{+0.22}_{-0.26}$

$P(m_1 \in M)$ $0.046^{+0.022}_{-0.016}$ | $P(q \in Q \mid m_1 \in M)$ $0.74^{+0.16}_{-0.22}$ | $P(q \in Q \mid m_1 \in M)$ $0.057^{+0.098}_{-0.046}$

$P(m_1 \in M)$ $0.0045^{+0.0061}_{-0.0026}$ | $P(q \in Q \mid m_1 \in M)$ $0.039^{+0.18}_{-0.036}$ | $P(q \in Q \mid m_1 \in M)$ $0.67^{+0.23}_{-0.38}$

$\chi_{\rm eff} = 0$, $\chi_{\rm eff} = 0.2$

$\chi_p = 0.2$, $\chi_p = 0.5$

χ_p

$\chi_{\rm eff}$

$p(\chi_{\rm eff}, \chi_p \mid m_1, q)$